Efficient CIFAR-100 Modelling

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Abstract

We investigate deep learning models under strict parameter constraints on CIFAR-100. A depthwise separable CNN is proposed for classification under a 100k-parameter limit, achieving 57.7% test accuracy, while a residual-based GAN is used for image generation under a 1 million-parameter constraint, achieving an FID of 46.3. The results show that competitive performance can be supported through careful architectural design despite severe parameter and optimisation step limitations.

Part 1: Classification

1 Methodology

We study discriminative multiclass classification on CIFAR-100 (60,000 32×32 RGB images; 100 classes) [7] under a strict budget of less than 100,000 trainable parameters and a minimum target test accuracy of 50%. These constraints motivate parameter-efficient convolutional architectures that preserve spatial inductive bias while reducing compute and capacity.

CNNs exploit spatial structure [8] via local receptive fields and weight sharing, providing a strong inductive bias for image recognition [8]. Standard convolution mixes spatial filtering and channel interactions with cost $\mathcal{O}_{\text{standard}}$ (Eq. 1), which becomes prohibitive under a 100k-parameter budget. Here, H and W denote spatial dimensions, C_{in} and C_{out} the number of input and output channels, and k the kernel size.

$$\mathcal{O}_{\text{standard}} = HWC_{\text{in}}C_{\text{out}}k^2 \tag{1}$$

We therefore use depthwise separable convolutions, reducing complexity to $\mathcal{O}_{\text{DW}}$ (Eq. 2), enabling deeper feature hierarchies within the constraint [1]. Cross-entropy (Eq. 3) is used for training.

$$\mathcal{O}_{\text{DW}} = HW\left(C_{\text{in}}k^2 + C_{\text{in}}C_{\text{out}}\right)$$

$$\mathcal{L}_{\text{CE}} = -\sum_{c=1}^{C} y_c \log(\hat{p}_c) \tag{2}$$

To improve optimisation stability under limited steps, we add residual connections within depthwise blocks [3] and apply Batch Normalisation after convolutions [5]. ReLU and SiLU were compared where ReLU was selected due to comparable test accuracy with lower overhead. Channel width increases as spatial resolution decreases via strided depthwise convolutions, and global average pooling replaces fully connected layers to reduce parameters and encourage global discriminative features [10].

Figure 1 shows a heatmap where there is a clear performance peak at higher learning rates (0.006–0.007) combined with moderate weight decay, indicating that the selected hyperparameters lie within a stable, high-performing region of the search space. The values for label smoothing, dropout rate and warmup steps were chosen through rigorous multi-grid search. 216 combinations were evaluated to get the highest validation accuracy of 59%.

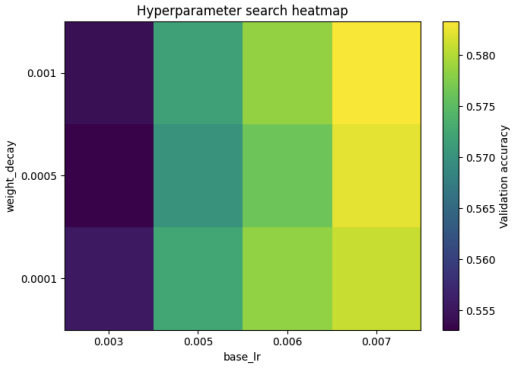


Figure 1: Validation accuracy heatmap for learning rate and weight decay with label smoothing=0.2, dropout=0.1, and warm-up steps fixed

2 Results

The final classifier contained 99,416 parameters and is evaluated over 10,000 optimisation steps as shown in Figure 2. Both metrics increase steadily in the early stages of training, indicating effective feature learning and stable optimisation. The model reaches a final training accuracy of 69.0%±6.0% and a test accuracy of 57.7%±6.3%. Train set performed 11.3% better than the test set. During the initial training phase, test accuracy is briefly above the training accuracy, which is caused by regularisation effects and learning rate warm-up, suppressing early training performance. As optimisation progresses beyond 2,000 steps, training accuracy surpasses test accuracy as the model increasingly fits the training distribution. Training accuracy continues to improve throughout optimisation, while test accuracy begins to plateau after 6,000 steps. This suggests diminishing returns beyond this point under the fixed model capacity. This divergence indicates moderate overfitting, reflecting the limited representational capacity imposed by the strict parameter and optimisation constraints. Despite this, the model demonstrates stable convergence and achieves competitive generalisation performance within the imposed limits. Variance across runs is non-negligible, overlapping standard deviations and consistent convergence indicate that the model does not exhibit high-variance or unstable training.

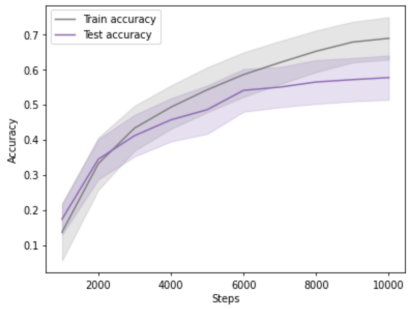


Figure 2: Training and test accuracy of classifier over 10,000 optimisation steps

3 Limitations

Despite achieving competitive performance under strict parameters and optimisation constraints, the proposed classifier exhibits several limitations. Limited accuracy is correlated with misclassifications frequently occurring between visually similar or semantically related classes, reflecting the difficulty of fine-grained discrimination at the 32×32 resolution of the dataset. While the model captures global shape and texture cues, it struggles to distinguish

subtle inter-class differences, particularly for categories with overlapping visual features. The moderate overfitting caused by the restricted representational capacity from the parameter limit represents a bottleneck rather than unstable optimisation. The strict 100,000 parameter limit with the use of depthwise separable convolutions constrains cross-channel interactions and limits the model's ability to learn fine-grained semantic distinctions. Despite architectural choices and regularisation mitigating extreme overfitting, the reduced expressive capacity biases the model towards higher-level, globally discriminative features such as general shapes and textures, at the expense of detailed semantic separation. These results highlight the inherent trade-off between computational efficiency and precise classification accuracy under strict resource constraints. The observed plateau in test accuracy suggests that increasing optimisation steps would lead to diminishing returns under fixed model capacity. This indicates that further improvements in generalisation would be more effectively achieved through increased representational capacity by increasing the number of parameters rather than further optimisation steps, although such changes fall outside the imposed constraints.

Part 2: Generative model

4 Methodology

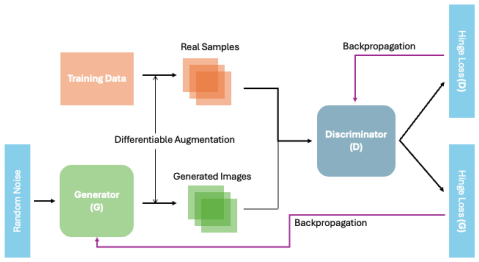


Figure 3: Residual GAN Architecture Diagram

The objective of this task is to learn an unconditional generative model capable of sampling realistic and diverse images from the CIFAR-100 data distribution under 1 million parameters and 50,000 optimisation steps.

A generative adversarial network (GAN) is well suited to this task as it enables direct learning of complex, high-dimensional image distributions [2], while residual architectures improve optimisation stability and sample quality under limited model capacity [11]. The generator–discriminator architecture is shown in Figure 3. The generator G maps a latent vector $z \sim \mathcal{N}(0, I)$ to an image $G(z) \in \mathbb{R}^{3\times32\times32}$. A fully connected projection reshapes z into a 4×4 feature map, followed by three residual upsampling blocks (4→8→16→32). A final same-resolution residual block refines spatial detail before a tanh-activated output layer. Batch normalisation and ReLU activations are used throughout the generator.

The discriminator D consists of an initial convolution, three residual downsampling blocks (32→16→8→4), and a linear output layer producing a real/fake score. Batch normalisation is omitted in D to improve training stability. The combined model contains 822,324 parameters.

Training uses the hinge loss formulation [9]:

$$\mathcal{L}_D = \mathbb{E}_{x\sim p_{\text{data}}}[\max(0, 1 - D(x))] + \mathbb{E}_{z\sim p(z)}[\max(0, 1 + D(G(z)))] \tag{3}$$

$$\mathcal{L}_G = -\mathbb{E}_{z\sim p(z)}[D(G(z))]. \tag{4}$$

Both networks are optimised using Adam with equal learning rates of 0.0003 and a one-to-one update ratio. Learning rates are decayed by a factor of 0.5 after 40,000 optimisation steps. Differentiable data augmentation (colour perturbations and small spatial translations) is ap-

plied to both real and generated images, with augmentation strength scheduled from strong early regularisation to weaker late-stage sharpening [12]. An exponential moving average (EMA) of generator parameters is maintained and used for sampling and evaluation [6]. Model performance is assessed using the Fréchet Inception Distance (FID) [4] computed on the CIFAR-100 test set using the EMA-averaged generator, alongside qualitative inspection of generated samples and latent interpolations.

5 Results

While abiding the computational constraints, the final model achieved an FID of 46.3, indicating moderate sample realism for unconditional CIFAR-100 generation. As shown in Figure 4, generated images are not excessively blurry and show coherent global structure, consistent colour distributions, and plausible textures. While object identity is occasionally vague, this is expected given the absence of class conditioning, many samples display realistic shapes and spatial organisation rather than noise or collapsed modes. Sample diversity is illustrated by the batch in Figure 4 and further assessed through latent space interpolations in Figure 5. Interpolations between randomly sampled latent vectors produce transitions that are midpoint realistic, rather than collapsing into linear alpha blends or blurred averages. This indicates that the generator has learned a continuous but imperfect latent manifold. Samples within a batch are visually diverse and do not converge to a small set of modes, suggesting no evidence of mode collapse. Although the model does not fully capture all fine-grained CIFAR-100 categories, it represents a broad subset of the dataset distribution under the given parameter and optimisation limits. Nearest-neighbour analysis in pixel space yields a mean L2 distance of 9.47±1.95 between generated samples and training images, suggesting that samples are not direct memorisations of individual training examples.

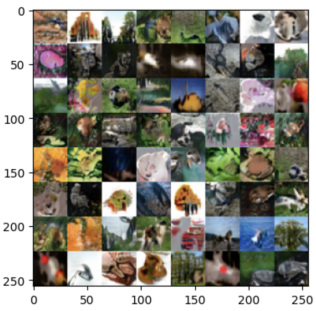


Figure 4: 64 random sample generated images

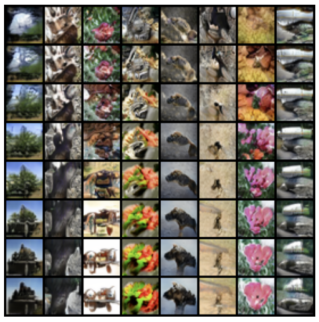


Figure 5: latent interpolations between randomly sampled latent vectors

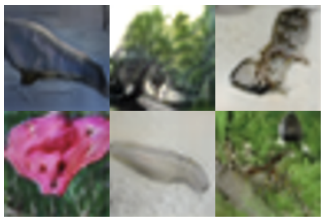


Figure 6: Handpicked generated images of a Seal, Truck, Kitten, Flower, Worm, Butterfly

6 Limitations

Due to the strict problem constraints, the proposed model shows several limitations. The final FID score of 46.3 reflects the difficulty of unconditional generation on CIFAR-100, where there is a large number of visually diverse classes must be modelled without class conditioning. As a result, some generated samples exhibit ambiguous object identity and reduced semantic clarity. The parameter and optimisation step limit restricts the model's capacity. While residual connections and regularisation improve stability, they limit the generator's ability to capture fine-grained details and high-frequency structure. Adversarial training is still sensitive to optimisation dynamics. Despite using learning-rate decay, EMA smoothing, and differentiable augmentation to improve convergence, some variability across runs still persists, particularly during the later training stages. While FID measures global distributional similarity, it does not measure perceptual realism or class-specific fidelity, encouraging complementary qualitative evaluation using generated samples and latent interpolations.

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